

医药翻译PRD.pdf - 双语对照预览

原页 1 · 原文

医药翻译PRD

原型参考文件持续更新中<http://10.3.75.127:3000/>

一、项目背景

1.1 业务背景

医药行业在日常研发、注册申报、学术交流及国际化合作中，涉及大量跨语言文档处理需求。典型场景包括：

- 药品临床试验报告、研究者手册、知情同意书等中英文文档互译；
- 医疗器械技术文件、说明书、标签的多语言版本制作；

PA 申报资料的快速翻译与对照审核。

- 医药学术论文、FDA/EMA/ NMPA

目前，用户主要依赖通用翻译工具或人工翻译。通用工具缺乏医药专业术语库支持，导致关键术语翻

译不准确、前后不一致；人工翻译成本高、周期长，无法满足高频、大批量的实时翻译需求。此外，现有工具对 Word、PDF 文档的格式保留能力不足，翻译后需大量人工调整排版，严重影响效率（e.g., FDA, EMA, NMPA）。

1.2 用户痛点

痛点 描述

术语不准确

通用翻译引擎无法识别“随机对照试验”“主要终点”“不良事件”等医药专业术语，翻译结果不可用

格式易破坏

PDF 或 Word 翻译后，表格、页眉页脚、图表布局错乱，需要大量人工修复

缺乏对照版本

翻译后仅有译文，无法快速与原文对照审核，双语版本需人工拼接

历史无法追溯

每次翻译需重新上传、设置语言，无法查阅之前的翻译结果

原页 1 · 译文

Medical Translation PRD

Prototype reference document is continuously being updated at <http://10.3.75.127:3000/>

I. Project Background

The pharmaceutical industry frequently deals with

multilingual document processing in daily R&D, regulatory submissions, academic exchanges, and international collaborations. Typical scenarios include:

- Bilingual translation of clinical trial reports, investigator brochures, and informed consent forms;
- Creation of multilingual versions for medical device

Currently, users mainly rely on general-purpose translation tools or manual translation. General tools lack specialized and inconsistent translations of key terms. Manual translation is costly and time-consuming, failing to meet the demand for existing tools poorly preserve formatting in Word and PDF documents—requiring extensive manual reformatting after translation, severely impacting efficiency.

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and inconsistent translations of key terms. Manual translation is costly and time-consuming, failing to meet the demand for

existing tools poorly preserve formatting in Word and PDF documents—requiring extensive manual reformatting after translation, severely impacting efficiency.

1.2 User Pain Points

| Pain Point | Description |

|-----|-----|

| Inaccurate Terminology | General translation engines fail to recognize pharmaceutical terms like "randomized controlled trial", "primary endpoint", and "adverse event", resulting in unusable translations. |

| Formatting Easily Broken | After translating PDF or Word files, tables, headers/footers, and figure layouts become disordered, requiring significant manual correction. |

| Lack of Parallel Versions | Only translated text is provided without a quick way to compare it against the original; bilingual versions must be manually assembled. |

| No Historical Traceability | Each translation requires re-uploading and re-setting language preferences, making it impossible to retrieve previous translation results. |

多语言配置繁琐

源语言、目标语言每次手动选择，效率低下

数据安全顾虑

医药文档涉及敏感信息，使用公共翻译工具存在数据泄露风险

多文件处理不便

一次性上传多个文档后，缺乏统一管理和对比预览能力

1.3 技术可行性

• 机器翻译 + 医药术语库的混合模式已在行业内成熟应用，可保证术语准确率。

• 文档格式保持技术（如 Apache POI、Aspose、PDF 解析引擎）可实现 Word/PDF 的版面稳定输出；

• 单点登录与数据隔离方案（SSO + 用户空间）可满足企业级安全要求。

二、产品价值

2.1 用户价值

价值点 说明

术语准确统一

内置医药通用术语库，确保同一术语全文一致、行业标准规范，可直接用于正式文档

格式零调整

翻译后 Word/PDF 保持原排版（允许长度微调，方向、结构不变），无需人工修复

效率大幅提升

文本翻译即时响应，文件翻译批量处理，历史记录一键复用，平均翻译时间缩短 80% 以上

对照审核便捷

文件翻译自动生成左右对照预览，并提供纯译文版和双语对照版下载，满足不同场景需求

历史可追溯

所有翻译任务自动保存，支持搜索、筛选、快速复现，避免重复工作

数据安全隔离

单点登录 + 用户数据隔离，保障医药文档的保密性和合规性

Multilingual configuration is cumbersome—source and target languages must be manually selected each time, resulting in low efficiency.

Data security concerns—pharmaceutical documents contain sensitive information, and using public translation tools poses a risk of data leakage.

Inconvenience in handling multiple files—after uploading several documents at once, there is no unified management or side-by-side preview capability.

1.3 Technical Feasibility

• Document format preservation technologies (such as Apache POI, Aspose, and PDF parsing engines) can achieve stable layout output for Word/PDF files.

• Single Sign-On (SSO) combined with user-space isolation ensures uniformity of terms throughout the document and compliance with industry standards—ready for use in official documents.

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• Single Sign-On (SSO) combined with user-space isolation ensures uniformity of terms throughout the document and compliance with industry standards—ready for use in official documents.

2. Product Value

2.1 User Value

Accurate and consistent terminology | Built-in pharmaceutical terminology database ensures uniformity of terms throughout the document and compliance with industry standards—ready for use in official documents.

Stable layout output | Translated Word/PDF files retain original layout (allowing minor length adjustments while preserving direction and structure)—no manual adjustments required.

Significant efficiency improvement | Instant response for text translation; batch processing for files; one-click reuse of historical records—average translation time reduced by over 80%.

Convenient comparison and review | Automatically generates side-by-side preview for translated documents, offering both pure translation and bilingual comparison versions for download to meet different scenarios.

Full traceability | All translation tasks are automatically saved, supporting search, filtering, and quick re-execution—preventing redundant work.

Data security and isolation | SSO + user data isolation ensures confidentiality and compliance of pharmaceutical documents.

原页 3 · 原文

操作直观

任务导向型界面，一目了然，无需培训即可上手

2.2 业务价值

价值点 说明

降低翻译成本

机器翻译取代大部分人工翻译，减少外包开支，预估年度成本降低 60%–70%

加速业务流转

即时翻译能力使药品注册资料、临床试验文档的处理周期从数天缩短至分钟级

提升国际竞争力

快速生成高质量多语言文档，助力企业拓展海外市场

知识资产沉淀

历史记录形成企业医药翻译语料库，可反哺术语库优化和未来 AI 训练

三、需求

3.1 用户认证与数据隔离

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作为用户,我想要系统优先支持单点登录(如飞书),以便使用企业统一身份认证,无需记忆多套账号密码,提升登录效率。

1 系统登录页应优先展示“飞书登录”按钮，用户点击后跳转至飞书授权页面。

2 若飞书 SSO 因配置不可用，系统应提供备用的账号密码登录表单。

3 使用任一方式登录成功后，用户进入系统主界面，且会话保持有效直到手动退出或超时。

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作为用户,我想要系统实现数据隔离,每个用户只能查看和操作自己的翻译记录,以便保障数据安全与隐私。

验收条件：

原页 3 · 译文

Intuitive operation — Task-oriented interface, clear at a glance, no training required.

2.2 Business Value

Value Point | Description

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Reduce translation costs | Machine translation replaces most manual translation, reducing outsourcing expenses; estimated annual cost reduction of 60% – 70%.

Accelerate business processes | Real-time translation capability shortens processing time for drug registration documents and clinical trial documents from days to minutes.

Enhance international competitiveness | Rapid generation of high-quality multilingual documents supports enterprise expansion into overseas markets.

Knowledge asset accumulation | Historical records form an enterprise-specific translation corpus, which can improve terminology databases and support future AI training.

Acceptance Criteria:
1. The system login page should prominently display a "Feishu Login" button; clicking it redirects users to the Feishu authorization page.
2. If Feishu SSO is unavailable due to configuration issues, the system must provide a backup username/password login form.
3. After successful login via either method, the user enters the main system interface with an active session that persists until manually logged out or timed out.

• As a user, I want the system to prioritize Single Sign-On (SSO) using Feishu for authentication, improving login efficiency without remembering multiple account credentials.

Acceptance Criteria:

1. The system login page should prominently display a "Feishu Login" button; clicking it redirects users to the Feishu authorization page.
2. If Feishu SSO is unavailable due to configuration issues, the system must provide a backup username/password login form.
3. After successful login via either method, the user enters the main system interface with an active session that persists until manually logged out or timed out.

• As a user, I want the system to implement data isolation so that each user can only view and operate their own translation records, ensuring data security and privacy.

Acceptance Criteria:

原页 4 · 原文

1 用户 A 登录后，左侧历史记录列表中只显示用户 A 的翻译任务，不显示其他用户的任何记录。

2 用户 A 无法通过直接输入 URL 或其他方式访问用户 B 的翻译结果或下载文件。

3 系统应基于登录用户身份进行所有数据操作的后台校验，确保数据隔离的安全性。

3.2 通用语言配置

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原页 4 · 译文

A1. After User A logs in, the left history list displays only User A's translation tasks and shows no records from other users.
2. User A cannot access User B's translation results by directly entering a URL or any other method.
3. The system shall perform backend validation based on the logged-in user's identity to ensure data isolation security.

2. The system shall ensure data isolation security by performing backend validation based on the logged-in user's identity, preventing User A from accessing User B's translation results or downloading files through direct URL entry or other methods.

3. The system shall perform backend validation based on the

作为用户,我想要系统至少支持中文与英文互译,并尽量覆盖其他常见语种(如日语、韩语、德语、法语等),以便应对多语种翻译场景

验收条件:

- 1 在文本翻译或文件翻译模式的输入区域上方,应提供源语言和目标语言两个下拉选择框,用户可自由选择任一支持的语言。
 - 2 源语言下拉框中应包含“自动检测””中文“”英文“选项,且默认为“自动检测”(选项,当用户输入中文文本时,系统应自动识别为“中文”;输入英文文本时自动识别为“英文”,并在下拉框中显示识别结果。文件上传同理
 - 3 当用户手动将源语言改为特定语言(例如“英语”)后,系统应以用户手动选择的语言为准进行翻译,不再自动覆盖。
 - 4 目标语言下拉框不提供自动检测,用户必须手动选择目标语言。枚举值至少包含“中文”“英文”
- 3.3 医药行业通用术语库挂载

• 作为用户,我想要系统在翻译过程中调用医药行业通用术语库,以便确保专业术语翻译准确

验收条件:

- 1 当用户提交翻译请求(文本或文件)后,系统后台必须调用医药行业通用术语库进行术语匹配与替换。
- 2 对于术语库中包含的医学术语(如“随机对照试验”(如“random ized controlled trial”),不得出现非术语库的直译。

As a user, I want the system to support bidirectional translation between Chinese and English, and as many other common languages (e.g., Japanese, Korean, German, French, etc.) to ensure data isolation security.

3.2 General Language Configuration

Acceptance Criteria:

1. The system shall provide two dropdown menus for source language and target language selection. The source language dropdown must include an “Auto Detect” option, and the default value should be “Auto Detect”. When the user inputs Chinese text, the system should automatically recognize it as “Chinese”; when inputting English text, it should automatically recognize it as “English”, and display the recognized result in the dropdown. The same applies to file uploads.

Acceptance Criteria:

1. The system shall use the language selected by the user as the standard for translation, and no longer override it automatically.
2. The source language dropdown must include options “Auto Detect”, “Chinese”, and “English”, with “Auto Detect” as the default. When the user inputs Chinese text, the system should automatically recognize it as “Chinese”; when inputting English text, it should automatically recognize it as “English”, and display the recognized result in the dropdown. The same applies to file uploads.
3. If the user manually changes the source language to a specific language (e.g., “English”), the system shall use the user-selected language for translation and no longer override it automatically.
4. The target language dropdown does not support auto-detection; the user must manually select the target language. The enumeration must include at least “Chinese” and “English.”

3.3 Medical Industry General Terminology Library Mounting

• As a user, I want the system to call the medical industry general terminology library during translation to ensure accurate and consistent professional term translations.

Acceptance Criteria:

1. Upon submission of a translation request (text or file), the system backend must invoke the medical industry general terminology library for term matching and replacement.
2. For medical terms included in the terminology library (e.g., “randomized controlled trial”), the translation must uniformly use the standardized translation from the library (e.g., “randomized controlled trial”)—no non-library direct translations are allowed.

- 3 若术语库中未收录某个词汇，系统应使用通用机器翻译引擎正常处理，不得因缺词而中断翻译。
- 3.4 通用翻译要求
- 作为用户,我想要点击发送后,系统显示翻译状态提示(加载中或成功或失败),以便了解翻译进度,失败时支持重试
- 验收条件：
- 1 用户点击“发送”后，发送按钮应立即置灰，进入翻页结果页面，并显示“翻译中”或加载动画（防止重复点击。
- 2 翻译成功完成后，页面左侧展示原文，页面右侧展示译文，并显示“翻译成功”提示（可短暂显示后自动消失）。
- 3 若因网络或服务器原因翻译失败，应显示明确的“翻译失败”提示，并在译文区域展示“重试”按钮。用户点击重试后，系统重新发起翻译请求。
- 4 重试成功后，状态应更新为成功；重试仍失败，可再次显示重试按钮，不限制重试次数。
- 3.3.1 文本翻译
- 作为用户,我想要在文本翻译模式下输入待翻译的医药文本,以便快速完成短文本或片段的翻译任务
- 验收条件：
- 1 当用户处于文本翻译模式时，右侧应显示多行文本输入框，用户可在此输入最大字符长度为5000的医药文本。
- 2 用户输入文本后点击“发送”按钮，系统应立即开始翻译处理。

3. If a term is not in the terminology database, the system shall use a general machine translation engine to process it normally, without interrupting the translation due to missing terms.
- 3.4 General Translation Requirements
- As a user, I want to see the translation or loading animation (e.g., "Loading", "Success", or "Failure") after clicking "Send", so I can understand the progress of the translation. If a failure occurs, retry functionality should be supported.
1. After the user clicks "Send", the button should immediately become disabled, and the result page, and display either "Translating..." or a loading animation to prevent repeated clicks.
2. Upon successful translation completion, the original text can be displayed on the left side, the translated text on the right side, with a "Translation Successful" message shown briefly before automatically disappearing.
3. If translation fails due to network or server issues, an explicit "Translation Failed" message should be displayed, along with a "Retry" button in the translation area. When the user clicks "Retry", the system should resend the translation request.
4. After a successful retry, the status should update to "Success"; if the retry still fails, the "Retry" button should reappear, with no limit on the number of retries.
- 3.3.1 Text Translation
- As a user, I want to input medical text in text translation mode so that I can quickly complete short texts or segments' translation tasks.
- Acceptance Criteria:
1. When the user is in text translation mode, a multi-line text input box should appear on the right side, allowing the user to enter medical text up to a maximum of 5000 characters.
2. After the user inputs the text and clicks the "Send" button, the system should immediately begin processing the translation.

- 3 进入翻译处理过程中，“发送”按钮应立即置灰，并显示“翻译中”或加载动画，直到翻译结束后（成功或者失败），按钮恢复可用
- 作为用户,我想要翻译结果直接显示在页面右侧(非对话气泡),以便清晰查看原文、译文内容,对译文内容进行复制
- 验收条件：
- 1 文本翻译完成后，右侧页面不应出现对话气泡样式，而应采用清晰的卡片或区域分别展示“原文”和“译文”。

3. During the translation process, the "Send" button should immediately be disabled and display "Translating..." or a loading animation, and the button becomes usable again after the translation is complete (whether successful or failed), at which point the button becomes usable again.
- As a user, I want the translation results to appear directly on the right side of the page (not in a tooltip bubble) so that I can clearly view the original text, the translated text, and copy the translated content.
- Acceptance Criteria:
1. After the text translation is completed, the right side of the page should not display a tooltip-style bubble, but should use a clear card or area to separately display the "Original Text" and the "Translated Text".

- 2 原文和译文应明确标识（如标题“原文”、“译文”），以便用户对此区分。
 - 3 在译文展示区域旁边应提供一个“复制”按钮或图标。复制翻译内容。
 - 4 用户点击复制按钮后，译文内容被复制到系统剪贴板，并出现“已复制”的短暂提示。
 - 5 复制的内容应完整且格式为纯文本，无多余空格或换行。
- 作为用户，我想要文本翻译仅支持单轮问答，不维持多轮对话上下文，以便聚焦单次翻译任务，避免干扰。
- 验收条件：
- 1 用户完成一次文本翻译后，若再次输入新文本并点击发送，系统应将此次翻译视为一个全新的独立任务，不与之前的原文或译文产生上下文关联（例如不会自动指代前文的宾语）。
 - 2 每次翻译的结果独立展示在右侧，不叠加显示历史翻译记录，历史记录在左侧列表中另行保存。
- 3.3.2 文件翻译

- clearly label the original text and the translated text, and provide a "Copy" button or icon next to the translation content.
- After the user clicks the copy button, the translated text should be copied to the system clipboard, followed by a brief prompt saying "Copied."
- The copied content must be complete, in plain text format, without extra spaces or line breaks.
- As a user, I want text translation to support only single-turn queries without maintaining multi-turn conversation context, so that I can focus on individual translation tasks without interference.
- Acceptance Criteria:
- 1. After completing one text translation, if the user enters new text and clicks send again, the system should treat this as a completely new and independent task, with no contextual connection to previous original or translated text (e.g., it should not automatically refer to the subject of the previous sentence).
 - 2. Each translation result should be displayed independently on the right side, without stacking or showing historical translations (historical records are saved separately in the left-side list).
- 3.3.2 File Translation

3.4 结果展示与下载

3.4 Results Display and Download

- 3.5 历史记录管理
- 3.6 界面与体验

- 3.5 Historical Record Management
- 3.6 Interface and Experience

3.3 非功能性目标

维度 目标

性能

文本翻译响应 ≤ 2 秒；文件翻译响应 ≤ 30 秒 (10 页以内)

可用性 系统全年可用性 ≥ 99.5%

安全性

数据加密存储，传输使用 HTTPS，用户会话隔离

兼容性 支持 Chrom

e、Edge 最新两个版本，分辨率 1280×

720 及以上

可维护性 术语库可独立更新，无需发版

3.3 Non-functional Objectives

Dimension | Objective

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Performance | Text translation response ≤ 2 seconds; File translation response ≤ 30 seconds (for documents up to 10 pages)

Availability | System annual availability ≥ 99.5%

Security | Data encrypted at rest and in transit using HTTPS; user sessions isolated

Compatibility | Supports latest two versions of Chrome and Edge; resolution 1280×720 or higher

Maintainability | Terminology database can be updated independently without releasing a new version